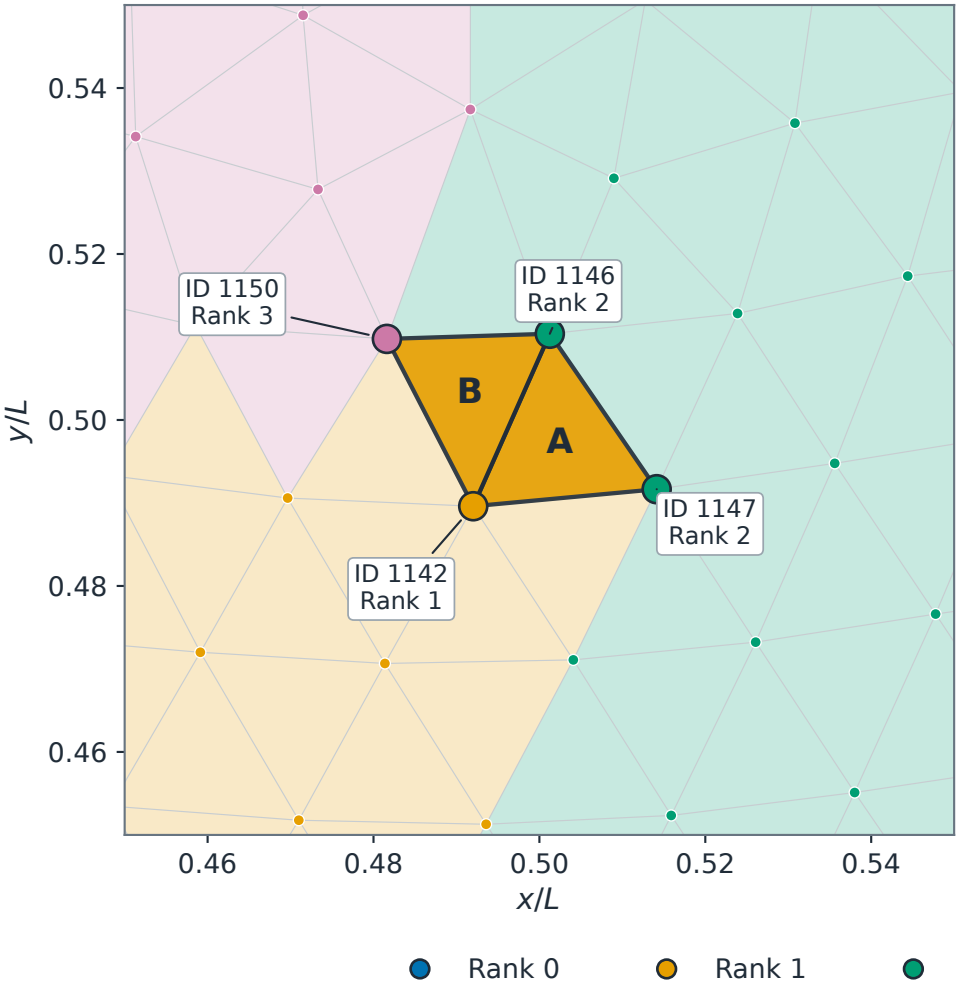
# Cross-rank triangles: two examples from the native mesh

## (a) Native triangles at a rank boundary




## (b) Responsibility and routing

Triangle fill: computing rank
Node colour: primary owning rank

### A | Vertex ranks: 1, 2, 2

Smallest ID: 1142, owned by rank 1.
Rank 1 computes this triangle.
The two rank-2 contributions
are sent from rank 1 to rank 2.

### B | Vertex ranks: 1, 2, 3

Smallest ID: 1142, owned by rank 1.
Rank 1 computes this triangle.
Contributions for IDs 1146 and 1150
are sent to ranks 2 and 3.

*Only the instance containing the winning primary node claims the triangle.*

Ownership rule: select the smallest ParticleID among the three vertices;
its primary owning rank evaluates the residual and distributes one contribution to each vertex.

Static full-mesh example, equal timesteps. Geometry and MPI packets are captured from the solver.